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ORIGINAL ARTICLE

Derivation of the Sagnac (Earth-rotation) correction and analysis of its accuracy for GNSS applications

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Abstract
Global Navigation Satellite Systems (GNSS) applications require computation of the geometric range between the satellite vehicle at the time-of-signal transmission and the receiver antenna location at the time-of-signal reception. This computation requires attention to the frames of reference due to the rotation of the Earth-Centered Earth-Fixed (ECEF) frame during the time-of-signal propagation. Three range computation approaches are commonplace and will be discussed herein. The first is the Global Positioning System Interface Control Document recommendation to rotate the ECEF frames to a common reference time. The other two are forms of the Sagnac correction. The Sagnac derivations already in the literature are either limited to stationary receivers or lack the connection between the Earth-centered inertial (ECI) and ECEF frames. Neither form of the Sagnac correction exactly reproduces the geometric range. They are approximations. The literature does not currently contain an analysis of the error involved in using either form of the Sagnac correction. This article makes two contributions: (1) it presents derivations for both forms of the Sagnac correction that are valid for moving receivers and that maintain the connection between the ECI and ECEF frames; and (2) it analyzes the error of the Sagnac correction for orbits of different radius. The analysis shows that Sagnac corrections introduce range errors less than 7.57×10^{-4} meters for GNSS satellites at medium Earth orbit.

Keywords Sagnac · GNSS · Earth rotation · Satellite

1 Introduction

A Global Navigation Satellite System (GNSS) satellite transmits its signals at the time-of-signal transmission. The time-of-signal reception at a receiver is later due to a nonzero time-of-signal propagation. Due to Earth's rotation, the Earth-Centered Earth-Fixed (ECEF) frame rotates during the time-of-signal propagation. This results in a distinct orientation of the ECEF frames at the time-of-signal transmission and reception. Therefore, Earth rotation correction is a necessary computational step for all GNSS satellite positioning applications (Hu et al. 2022; Farrell 2008; Groves 2013; Teunissen and Montenbruck 2017; Misra and Enge 2011). Without correcting the Earth rotation effects, the satellite position can be in error by over 30 meters (Ashby 2004). This rotation compensation is often referred to as the Sagnac correction in GNSS applications.

The Global Positioning System (GPS) Interface Control Document (ICD) recommends that each satellite position vector should be rotated into the ECEF frame at the time-of-signal reception (see Eq. (6) herein). Nonetheless, practitioners often use approximations (see Eqs. (7)–(8)). Even though approximations should never be used without an accuracy assessment, to the authors' knowledge, there is no assessment in the literature of the accuracy of these approximations. The main new contribution of this article is the analysis of the error incurred when using the approximate ECEF rotation correction methods.

The analysis requires an understanding of reference frames and notation capable of denoting the reference frame in which each position vector is represented. Related background is included in Appendix A. Section 2 presents background related to the topic of the Sagnac correction. Sec-

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tion 3 reviews the literature related to the Sagnac correction and discusses motivations for the Sagnac approximations. Section 4 derives preliminary results useful for the error analysis. Section 5 derives each approximation and analyzes the approximation error. Section 7 concludes the article.

2 Geometric range background

Because the Earth-centered inertial (ECI) frame is not rotating, it is straightforward to determine the geometric range in an ECI frame. Without loss of generality, define an ECI frame whose origin, axis, and velocity coincide with those of the ECEF frame of satellite s at the time-of-signal transmission denoted by t_s. The signal travels in this ECI frame from the satellite location[1] at that time $\mathbf{P}_s^I(t_s)$ to the receiver location $\mathbf{P}_r^I(t_r)$ at the time-of-signal reception denoted by t_r. The geometric range is

$$D = \|\mathbf{P}_r^I(t_r) - \mathbf{P}_s^I(t_s)\|. \tag{1}$$

This form of the range equation is correct, but not convenient because: (1) the user would like to know the receiver position in the ECEF frame; (2) the ICD provides equations for the computation of the satellite position at t_s in the ECEF frame at that time; and, (3) for signals from multiple satellites all received at t_r, each t_s is distinct.

Because norms are invariant under rotation, the geometric range can also be represented as either

$$D = \left\|\mathbf{R}_I^{E(t_r)}\left(\mathbf{P}_r^I(t_r) - \mathbf{P}_s^I(t_s)\right)\right\| \text{ or} \tag{2}$$

$$D = \left\|\mathbf{R}_I^{E(t_s)}\left(\mathbf{P}_r^I(t_r) - \mathbf{P}_s^I(t_s)\right)\right\| \tag{3}$$

where $\mathbf{R}_I^{E(t)}$ represents the rotation matrix from the ECI frame to the ECEF frame valid at time t.

Even though both $\mathbf{P}_r^I(t_r)$ and $\mathbf{P}_s^I(t_s)$ are well-defined and unique points, they will have different coordinates in the ECEF frame at different times, due to the rotation of the Earth axis with respect to the ECI frame axes. To keep track of these distinct coordinate vectors in Eqs. (2–3), define the following symbols:

- $\mathbf{P}_r^{E_r}(t_r) = \mathbf{R}_I^{E(t_r)}\mathbf{P}_r^I(t_r)$ is the position of the receiver at the time-of-signal reception represented in the ECEF frame at the time-of-signal reception.
- $\mathbf{P}_r^{E_s}(t_r) = \mathbf{R}_I^{E(t_s)}\mathbf{P}_r^I(t_r)$ is the position of the receiver at the time-of-signal reception represented in the ECEF frame at the time-of-signal transmission. This symbol is defined for completeness relative to the notation in Eqs. (2) and (3), but is not used further herein.
- $\mathbf{P}_s^{E_r}(t_s) = \mathbf{R}_I^{E(t_r)}\mathbf{P}_s^I(t_s)$ is the position of the satellite at the time-of-signal transmission represented in the ECEF frame at the time-of-signal reception.
- $\mathbf{P}_s^{E_s}(t_s) = \mathbf{R}_I^{E(t_s)}\mathbf{P}_s^I(t_s)$ is the position of the satellite at the time-of-signal transmission represented in the ECEF frame at the time-of-signal transmission.

This notation for the coordinate vector of a position is keeping track of a few things: the position of a specific point (i.e., $\mathbf{P}_r$ or $\mathbf{P}_s$) at a specific time (i.e., t_r and t_s), and the ECEF reference frame at a specific time (i.e., $E_r = E(t_r)$ or $E_s = E(t_s)$) (see Fig. 1). For further discussion, see Appendix A.

In addition, the following two symbols will subsequently become important:

- $\mathbf{P}_r^I(t_s)$ is the position of the receiver at the time-of-signal transmission in the ECI frame;
- $\mathbf{P}_r^{E_s}(t_s) = \mathbf{R}_I^{E(t_s)}\mathbf{P}_r^I(t_s)$ is the position of the receiver at the time-of-signal transmission represented in the ECEF frame at the time-of-signal transmission.

Using the notation of the first four bullets, Eq. (2) is equivalent to (4):

$$D = \left\|\mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_s^{E_r}(t_s)\right\|. \tag{4}$$

Equation (4) is theoretically equivalent to Eq. (1). The main difference is that Eq. (1) computes the position difference vector in I-frame, while Eq. (4) does so in the E_r-frame. Correct computation of the distance D using a norm, as in any of Eqs. (1)–(4), requires that the two vectors being subtracted must be represented in the same frame of reference. Equation (4) is more convenient than Eq. (1) because the receiver position $\mathbf{P}_r^{E_r}(t_r)$ is represented in and ECEF frame; however, it is not quite in a useful form because the ephemeris information and ICD describe computation of $\mathbf{P}_s^{E_s}(t_s)$, not $\mathbf{P}_s^{E_r}(t_s)$.

Using the third bullet above to solve for $\mathbf{P}_s^I(t_s)$ and substituting into the fourth bullet results in the following two relations:

$$\mathbf{P}_s^{E_s}(t_s) = \mathbf{R}_{E_r}^{E_s}\mathbf{P}_s^{E_r}(t_s) \text{ and } \mathbf{P}_s^{E_r}(t_s) = \mathbf{R}_{E_s}^{E_r}\mathbf{P}_s^{E_s}(t_s) \tag{5}$$

where $\mathbf{R}_{E_r}^{E_s} \triangleq \mathbf{R}_I^{E_s}\mathbf{R}_{E_r}^I$, and we have used the fact that for any rotation matrix its inverse is equal to its transpose. This rotation matrix will be explicitly defined later in the article (see Eq. (16) and Appendix A). The second expression in Eq. (5) is useful because the ICD provides equations for the computation of $\mathbf{P}_s^{E_s}(t_s)$, and the rotation matrix $\mathbf{R}_{E(t_r)}^{E(t_s)}$ can be computed using the time-of-signal propagation defined as $t_p = t_r - t_s$ (see Eqs. (A7) and (A17)). This allows Eq. (4)

[1] Appendix A provides additional information related to notation, reference frames, and the rotations between frames with coincident origins.

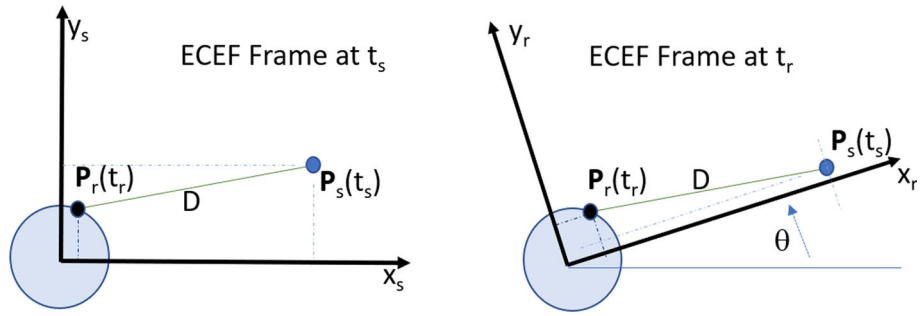

Fig. 1 Illustration of the ECEF frame x and y axes at times t_s and t_r. As stated before Eq. (1) and in Useful Fact A.1, the ECI frame is selected to be coincident with the ECEF frame at time t_s

to be manipulated, by substituting the expression for $\mathbf{P}_s^{E_r}(t_s)$ in Eq. (5) into Eq. (4), to provide the equation

$$D = \|\mathbf{P}_r^{E_r}(t_r) - \mathbf{R}_{E_s}^{E_r}\,\mathbf{P}_s^{E_s}(t_s)\|, \tag{6}$$

where both $\mathbf{R}_{E_s}^{E_r}$ and $\mathbf{P}_s^{E_s}(t_s)$ are known quantities. Therefore, the satellite position at a given time t_s has different representations in the ECEF frames valid at different times (e.g., t_s and t_r). This difference is solely attributable to Earth's rotation. The satellite's velocity is irrelevant.

Equation (6) is the method recommended in the GPS ICD. Even though the Earth rotation rate (i.e., $\omega_{ie} = 7.3 \times 10^{-5}$ radians per second) is small and the time-of-signal propagation satisfies $t_p < 0.1$ seconds, the vector $\mathbf{P}_s^{E_s}$ has norm on the order of 20×10^6 meters. Hence, the Earth rotation during the time-of-signal propagation cannot be neglected (Ashby 2004; Teunissen and Montenbruck 2017; Bidikar et al. 2016).

3 Sagnac literature and motivation

In applications, the effect of the Earth's rotation on the geometric range D is sometimes approximated using the Sagnac computation, which has been defined in two forms (Ashby 1995, 1999; Su 2001; Ashby 2004; Gift 2014):

$$D_s \triangleq ||\mathbf{P}_r^{E_s}(t_s) - \mathbf{P}_s^{E_s}(t_s)|| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_s}(t_s)\right) \tag{7}$$

$$D_r \triangleq ||\mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_s^{E_s}(t_s)|| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_r}(t_r)\right). \tag{8}$$

where c is the speed of light and $\boldsymbol{\omega}_e$ is the Earth rotation rate vector as defined in Eq. (A6). When the receiver is stationary, Sect. 4.1 shows that the right-hand sides of Eqs. (7) and (8) are equivalent (i.e., $D_s = D_r$). Sections 5.1 and 5.2, respectively, will derive D_s and D_r as approximations to D.

3.1 Caution with frames

Figure 1 shows the two points $\mathbf{P}_r(t_r)$ and $\mathbf{P}_s(t_s)$. The left image shows coordinate vectors for each point resolved along the ECI and ECEF E_s frame axes. The right image shows the same two points, with their coordinate vectors now resolved along the ECEF E_r frame axes. These two points are at the exact same positions in both images; however, because the ECEF axes have rotated from E_s to E_r, the coordinate vectors for both points (depicted by the dotted blue lines) are distinct in the two frames. Nonetheless, they have the same geometric range D between them: computed by Eq. (1) in the E_s frame (left image) and by Eq. (6) in the E_r frame (right image). Equation (6) has been derived from Eq. (1) without any approximations. Both equations take the norm of the difference vector between two position coordinate vectors resolved in the same frame.

Computation of distance by subtraction of vectors in different frames is theoretically incorrect and, in general, will not correctly compute the geometric range. Similarly, the cross-product is only correct for vectors defined in the same frame. Nonetheless, Eq. (8) is often used. A main contribution of this article is assessing its accuracy.

3.2 Sagnac literature

Equation (7) is motivated and derived in (Ashby 1995, 1999, 1994). That analysis starts (e.g., see Eq. (34) in (Ashby 1999)) from the geometric range D expressed in an ECI frame as in Eq. (1). In the final step of the derivation, the receiver is assumed to be stationary relative to the Earth frame, and then, the geometric range is computed in the ECEF frame at the time-of-signal transmission using the satellite and receiver positions at the time-of-signal transmission. Equation (7) presents the Sagnac correction in the form

$$\frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_s}(t_s)\right). \tag{9}$$

Subsequent research presented the geometric range in the form of Eq. (8), which represents the receiver position at t_r in the ECEF frame at t_r and the satellite position as t_s in the

E_s frame (see (Su 2001; Gift 2014; Takasu 2013), or Sects. 5.4.4 and 19.1.1 of (Teunissen and Montenbruck 2017)). In Eq. (8), the Sagnac correction has the form

$$S_r = \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_r}(t_r) \right). \tag{10}$$

Note that the vector operations of subtraction and cross-product in Eqs. (8) and (10) are performed between vectors in distinct frames (which is not proper); however, Eq. (8) provides a close approximation to D at the time-of-signal reception and is often used in practice. Defining the coordinates as $\mathbf{P}_r^{E_r}(t_r) = [a, b, u]^T$ and $\mathbf{P}_s^{E_s}(t_s) = [x, y, z]^T$, the Sagnac correction S_r is typically written as

$$S_r = \frac{\omega_{ie}}{c}(x\, b - a\, y), \tag{11}$$

which is a more computationally efficient implementation than Eq. (10) due to $\boldsymbol{\omega}_e$ containing two zero elements.

Derivation of D_r have also been present in (Hu and Farrell 2019; Leick et al. 2015) directly from Eq. (6) using a first-order approximation to $\mathbf{R}_z(\theta)$. These derivations are valid for moving receivers, because Eq. (6) delineates the rover position at the time-of-signal reception regardless of the receiver's motion status. Nonetheless, deriving Eq. (11) from Eq. (6) is unsatisfying for a few reasons. First, it lacks the connection between ECI and ECEF, which is important for physical insight. Second, neither reference analyzes a bound on the Sagnac approximation error. Hu and Farrell, (Hu and Farrell 2019) does define four components of the error, but analysis of their error magnitude is tedious, complex, and unpublished. Also, it fails to yield a notably smaller error bound than is presented herein. Therefore, analysis of the error between D and either D_s or D_r has not been presented in the current literature.

This article only discusses the Earth rotation compensation either by Eq. (6) or by the Sagnac approximations in Eqs. (7)–(8). Particularly, we focus on the error incurred when the approximations are used. Various relativistic effects are encountered in GNSS applications that are beyond the scope of this article. The interested reader is referred to (Ashby 2004, 1995; Post 1967).

3.3 Comparison of the methods

The expression for D_s is an approximation to D that is strictly valid only for a stationary receiver. A challenge to using the approximate geometric range D_s in GNSS applications is that many receivers acquire measurements from all satellites simultaneously at t_r; therefore, because each satellite has a distinct time-of-signal transmission, each satellite will have a distinct t_s, which results in each D_s being relevant to a distinct frame. Furthermore, for a non-stationary receiver, its ECEF position is distinct at each t_s and at t_r.

Both Eqs. (6) and (8) resolve positions in the E_r frame and use the $\mathbf{P}_s^{E_s}(t_s)$ vector that can be directly computed from the ephemeris data. The real comparison between them is the effect of the receiver clock error. The rotation matrix $\mathbf{R}_{E_s}^{E_r}$ is a function of $\theta = \omega_{ie}\, t_p$ (see Eq. (A16)) where t_p represents the true time-of-signal propagation. The receiver estimates the time-of-signal propagation by

$$\hat{t}_p = \tilde{t}_r - t_s$$

where $\tilde{t}_r = t_r + b_r$ is the measured time-of-signal reception and affected by unknown clock bias b_r. When a GNSS user computes the rotation matrix by $\mathbf{R}_{E_s}^{E_r}(\omega_{ie}\, \hat{t}_p)$, the effect of clock error is proportional to

$$\left\| \frac{\partial D}{\partial b_r} \right\| = \omega_{ie} \left\| \mathbf{P}_s^{E_s}(t_s) \right\| \approx 1460\, \frac{m}{s}.$$

Because the receiver clock error b_r can be significant, iteration is required as the receiver clock bias is estimated, to correctly compute the rotation matrix $\mathbf{R}_{E_s}^{E_r}$. In contrast, for the geometric range model using Eq. (8), the clock bias does not directly appear.

3.4 Contributions of this article

This article derives both forms of the Sagnac correction and analyzes the error between D and both D_s and D_r. The derivation for D_r for a non-stationary receiver and the quantification of the approximation errors are original to this article. Section 4 analyzes the relationship between D_r and D_s and presents notation and results for later use in the article. Section 5 derives both forms of the Sagnac correction. These derivations include analyses of the approximation errors made in the derivations. The analysis quantifies bounds on the error in the Sagnac computations, showing the error to be less than 7.57×10^{-4} meters for medium Earth orbit GNSS satellites.

4 Preliminary analysis

This section analyzes the relationship between D_r and D_s and defines notation and results that will be useful in the derivation and analysis of those two quantities.

4.1 Relation between D_s and D_r

The approximation D_s defined in Eq. (7) is related to D_r defined in (8). Starting from Eq. (8) by adding and subtracting $\mathbf{P}_r^{E_s}(t_s)$ within each term yields

$$
\begin{aligned}
D_r &= \left\| \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_r^{E_s}(t_s) + \mathbf{P}_r^{E_s}(t_s) - \mathbf{P}_s^{E_s}(t_s) \right\| \\
&\quad + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \left[\mathbf{P}_r^{E_s}(t_s) + \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_r^{E_s}(t_s) \right] \right) \\
&= \left\| \mathbf{P}_r^{E_s}(t_s) - \mathbf{P}_s^{E_s}(t_s) + \Delta\mathbf{P}^E \right\| \\
&\quad + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times (\mathbf{P}_r^{E_s}(t_s) + \Delta\mathbf{P}^E) \right), \qquad (12)
\end{aligned}
$$

where the right-hand side of Eq. (12) is a perturbation to D_s by

$$
\Delta\mathbf{P}^E = \Delta\mathbf{P}^E(t_r, t_s) = \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_r^{E_s}(t_s). \qquad (13)
$$

Using Eq. (A4) in Useful Fact A.3, the symbol $\Delta\mathbf{P}^E$ represents the relative change of the receiver's position coordinates in the ECEF frame during signal propagation. If the receiver is stationary in the ECEF frame, then $\mathbf{P}_r^{E_r}(t_r) = \mathbf{P}_r^{E_s}(t_s)$ and $\Delta\mathbf{P}^E = \mathbf{0}$; otherwise, $\Delta\mathbf{P}^E \neq \mathbf{0}$. Therefore, Eqs. (7) and (8) are equivalent only if the receiver is stationary in the ECEF frame.

This article conducts error analysis for both D_s and D_r. However, only D_r is universally applicable for general purposes, as it defines the receiver position $\mathbf{P}_r^{E_r}(t_r)$ at a common time t_r, valid for all GNSS signals received at t_r while transmitted at different t_s. Additionally, using D_r does not require knowledge of whether the receiver is stationary or in motion.

4.2 Motion of an Earth-fixed point in *I*-frame

Figure 2 illustrates an x-y planar cross section of the ECI I-frame. In the left portion of the figure, a receiver that is stationary in the Earth frame (i.e., it is Earth-fixed) is represented at t_s by point A (e.g., $\mathbf{P}_r^I(t_s)$). Due to the Earth's rotation, its ECI position $\mathbf{P}_r^I(t)$ changes from point A at time t_s to point $A' = \mathbf{P}_r^I(t_r)$ at t_r. The rotation angle of the Earth's surface relative to I-frame is denoted as $\theta = \omega_{ie}(t_r - t_s)$ as discussed in Eq. (A14). Using Eq. (A13) from Useful Fact A.4, the I-frame coordinates of an Earth-fixed point at times t_s and t_r are related by

$$
\mathbf{P}_r^I(t_r) = \mathbf{R}_z(-\theta)\,\mathbf{P}_r^I(t_s). \qquad (14)
$$

For future reference, define a new symbol

$$
\mathbf{P}_{r_0}^I(t_r) = \mathbf{R}_z(-\theta)\,\mathbf{P}_r^I(t_s). \qquad (15)
$$

The symbol $\mathbf{P}_{r_0}^I(t_r)$ is the I-frame location at t_r that corresponds to the receiver position at time t_s due only to the Earth-rotation through the angle θ. For an Earth-fixed receiver, $\mathbf{P}_r^I(t_r) = \mathbf{P}_{r_0}^I(t_r)$. For a receiver that is in motion relative to Earth, $\mathbf{P}_r^I(t_r) \neq \mathbf{P}_{r_0}^I(t_r)$.

4.3 Earth-relative position change

Consider the case where the receiver may be non-stationary in E-frame. The image on the right in Fig. 2 depicts this general scenario within the I-frame. The change in the receiver's ECI position from t_s to t_r is a result of both the Earth's rotation (A to A') and the receiver motion relative to the Earth (A' to B). Those points related to the receiver are defined as follows.

- The black point A represents the receiver's position at t_s, denoted as $\mathbf{P}_r^I(t_s)$. Its position coordinates in the E_s-frame are

  $$\mathbf{P}_r^{E_s}(t_s) = \mathbf{R}_I^{E_s}\,\mathbf{P}_r^I(t_s).$$

- The green point B represents the receiver's position at t_r, denoted as $\mathbf{P}_r^I(t_r)$.
- The blue point A' is $\mathbf{P}_{r_0}^I(t_r)$ as defined in Eq. (15). Its position coordinates in E_r-frame are

  $$\mathbf{P}_{r_0}^{E_r}(t_r) = \mathbf{R}_I^{E_r}\,\mathbf{P}_{r_0}^I(t_r).$$

  This is the position vector in the E_r-frame that a stationary receiver would have at t_r if it was at $\mathbf{P}_r^I(t_s)$ at t_s.

According to Part 1 of Useful Fact A.4, $\mathbf{P}_{r_0}^{E_r}(t_r) = \mathbf{P}_r^{E_s}(t_s)$.

The rotation matrices $\mathbf{R}_I^{E_s}$ and $\mathbf{R}_I^{E_r}$ are discussed in Part 4 of Useful Fact A.4. If at t_s, the ECI frame is defined to be coincident and aligned with the ECEF frame, then without loss of generality, $\mathbf{R}_I^{E_s} = \mathbf{I}$ and $\mathbf{R}_I^{E_r} = \mathbf{R}_z(\theta)$, where θ is defined in Eq. (A14). Therefore, because $\mathbf{R}_{E_r}^{E_s} = \mathbf{R}_I^{E_s}\,\mathbf{R}_{E_r}^I$:

$$
\mathbf{R}_{E_s}^{E_r} = \mathbf{R}_z(\theta) \text{and } \mathbf{R}_{E_r}^{E_s} = \mathbf{R}_z(-\theta). \qquad (16)
$$

The geometric distance D is independent of the receiver trajectory between times t_s and t_r; therefore, for the convenience of analysis that trajectory can be decomposed into two parts: (1) the circular arc due to the Earth rotation for an Earth-fixed point from $\mathbf{P}_r^I(t_s)$ to $\mathbf{P}_{r_0}^I(t_r)$ (i.e., A to A'); and (2) the Earth-relative motion from $\mathbf{P}_{r_0}^I(t_r)$ to $\mathbf{P}_r^I(t_r)$ (i.e., A' to B).

Using Eq. (15), the receiver's Earth-relative motion in the I-frame can be expressed as

$$
\begin{aligned}
\Delta\mathbf{P}_{r,r_0}^I &= \mathbf{P}_r^I(t_r) - \mathbf{P}_{r_0}^I(t_r) \qquad (17) \\
&= \mathbf{P}_r^I(t_r) - \mathbf{R}_z(-\theta)\,\mathbf{P}_r^I(t_s). \qquad (18)
\end{aligned}
$$

Using Part 1 of Useful Fact A.4, $\Delta\mathbf{P}_{r,r_0}^I$ is related to $\Delta\mathbf{P}^E$ defined in Eq. (13) as

$$
\Delta\mathbf{P}^E = \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_{r_0}^{E_r}(t_r) + \left(\mathbf{P}_{r_0}^{E_r}(t_r) - \mathbf{P}_r^{E_s}(t_s) \right) \qquad (19)
$$

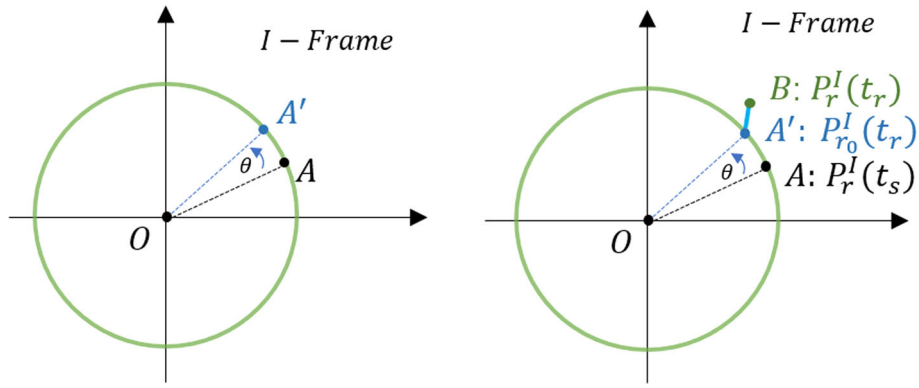

Fig. 2 Illustration of receiver motion in the I-frame. The green circle represents the Earth's surface viewed from above along its rotational axis. The Earth's surface is rotating counter-clockwise around the I-frame z-axis through the origin at the Earth's center. Left—A receiver that is stationary relative to the Earth will have its I-frame location rotate with the Earth's surface from point A to point A' through angle θ. Right—When a receiver is in motion relative to the Earth, its I-frame position B at time t_r can be decomposed into the component from A to A' due to the Earth-rotation and the component A' to B due to Earth-relative motion

$$
\begin{aligned}
&= \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_{r_0}^{E_r}(t_r) \\
&= \mathbf{R}_I^{E_r}\,\mathbf{R}_{E_r}^I\left(\mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_{r_0}^{E_r}(t_r)\right) \\
&= \mathbf{R}_I^{E_r}\left(\mathbf{P}_r^I(t_r) - \mathbf{P}_{r_0}^I(t_r)\right) \\
&= \mathbf{R}_z(\theta)\,\Delta\mathbf{P}_{r,r_0}^I,
\end{aligned} \tag{20}
$$

where we have used the facts that $\mathbf{P}_{r_0}^{E_r}(t_r) = \mathbf{P}_r^{E_s}(t_s)$ and $\mathbf{R}_I^{E_r} = \mathbf{R}_z(\theta)$.

5 Derivation of Sagnac

This section derives the Sagnac correction in the forms presented in Eqs. (7) and (8) and derives bounds on the magnitude of their approximation errors.

5.1 Derivation of D_s for stationary receivers

Assuming that the receiver is Earth-fixed (i.e., $\Delta\mathbf{P}^E = 0$), then using Eq. (14) and the expansion of Eq. (C21) of Appendix C,

$$
\mathbf{P}_r^I(t_r) = \mathbf{R}_z(-\theta)\,\mathbf{P}_r^I(t_s) \tag{21}
$$

$$
= \left(\mathbf{I} + \boldsymbol{\Omega}\,t_p + \frac{1}{2}\boldsymbol{\Omega}^2\,t_p^2\right)\mathbf{P}_r^I(t_s) \tag{22}
$$

$$
= \mathbf{P}_r^I(t_s) + t_p\,\boldsymbol{\Omega}\,\mathbf{P}_r^I(t_s) + \boldsymbol{\varepsilon}_1. \tag{23}
$$

In this expression, the $\boldsymbol{\Omega}\,\mathbf{P}_r^I(t_s)$ term is the tangential I-frame velocity and $\boldsymbol{\Omega}^2\,\mathbf{P}_r^I(t_s)$ is the centripetal acceleration of the receiver located at $\mathbf{P}_r^I(t_s)$; and, error $\boldsymbol{\varepsilon}_1 = \frac{1}{2}t_p^2\,\boldsymbol{\Omega}^2\,\mathbf{P}_r^I(t_s)$ has magnitude less than[2] 1.46×10^{-4} meters.

Substituting Eq. (23) into Eq. (1), the geometric range D from Eq. (1) can be expressed as

$$
D = \|\left(\mathbf{P}_r^I(t_s) + t_p\,\boldsymbol{\Omega}\,\mathbf{P}_r^I(t_s) + \varepsilon_1\right) - \mathbf{P}_s^I(t_s)\| \tag{24}
$$

$$
= \|\left(\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\right) + t_p\,\boldsymbol{\omega}_e \times \mathbf{P}_r^I(t_s) + \boldsymbol{\varepsilon}_1\|. \tag{25}
$$

Applying the first-order Taylor expansion as shown in Eq. (C23) of Appendix C, with $\mathbf{X} = \left(\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\right)$ and $\mathbf{Y} = \boldsymbol{\omega}_e \times \mathbf{P}_r^I(t_s)\,t_p + \boldsymbol{\varepsilon}_1$, to Eq. (25) yields

$$
\begin{aligned}
D &= \|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\| + \frac{\mathbf{X}\cdot\mathbf{Y}}{\|\mathbf{X}\|} \\
&\quad + \frac{1}{2}\frac{\mathbf{Y}\cdot\mathbf{Y}}{\|\mathbf{X}\|} + \varepsilon_4 \\
&= \|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\| \\
&\quad + \frac{\mathbf{X}\cdot\left(\boldsymbol{\omega}_e \times \mathbf{P}_r^I(t_s)\,t_p + \boldsymbol{\varepsilon}_1\right)}{\|\mathbf{X}\|} \\
&\quad + \frac{1}{2}\frac{\mathbf{Y}\cdot\mathbf{Y}}{\|\mathbf{X}\|} + \varepsilon_4 \\
&= \|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e\cdot\left(\mathbf{P}_r^I(t_s)\times\mathbf{X}\right)\,t_p}{\|\mathbf{X}\|} \\
&\quad + \frac{\mathbf{X}\cdot\boldsymbol{\varepsilon}_1}{\|\mathbf{X}\|} + \frac{1}{2}\frac{\|\mathbf{Y}\|^2}{\|\mathbf{X}\|} + \varepsilon_4
\end{aligned} \tag{26}
$$

[2] For the bounds in this section and the majority of the article, we use the following parameters for medium Earth orbit satellites with semi-major axis $\|\mathbf{P}_s\| = 27.0 \times 10^6 m$ (GPS Product Team 2020; BeiDou 2018; Galileo 2019): $\omega_{ie} = 7.3 \times 10^{-5}$rad/s, $t_p \in [\|\mathbf{X}\|_{\min}/c, \|\mathbf{X}\|_{\max}/c]$ s, $\|\mathbf{P}_r\| \approx 6.4 \times 10^6 m$. Different bounds apply to other categories of satellites, as will be discussed in Sect. 6.

$$= \|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right) t_p}{\|\mathbf{X}\|} + \frac{\mathbf{X} \cdot \boldsymbol{\varepsilon}_1}{\|\mathbf{X}\|} + \varepsilon_2 + \varepsilon_4 \tag{27}$$

where $\varepsilon_2 = \frac{1}{2}\frac{\|\mathbf{Y}\|^2}{\|\mathbf{X}\|}$ and $\|\mathbf{X}\|_{min} \leq \|\mathbf{X}\| \leq \|\mathbf{X}\|_{max}$ with[3]

$$\|\mathbf{X}\|_{min} = \|\mathbf{P}_s^I(t_s)\| - \|\mathbf{P}_r^I(t_s)\| = 20.60 \times 10^6 m \text{ and,}$$
$$\|\mathbf{X}\|_{max} = \sqrt{\|\mathbf{P}_r^I(t_s)\|^2 + \|\mathbf{P}_s^I(t_s)\|^2} = 27.75 \times 10^6 m.$$

The error term ε_4 represents high-order terms ($H.O.T$s) which have an upper bound of 4.54×10^{-5} meters (see Footnote 2. Defining the error term ε_s as

$$\varepsilon_s \triangleq \frac{\mathbf{X} \cdot \boldsymbol{\varepsilon}_1}{\|\mathbf{X}\|} + \varepsilon_2 + \varepsilon_3, \tag{28}$$

where

$$\begin{aligned}\varepsilon_3 &= \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right) t_p}{\|\mathbf{X}\|} - \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right) t_p}{c\, t_p} \\ &= \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right) t_p}{c\, t_p \|\mathbf{X}\|} \left(c\, t_p - \|\mathbf{X}\|\right) \\ &= \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right)}{c \|\mathbf{X}\|} \left(c\, t_p - \|\mathbf{X}\|\right),\end{aligned} \tag{29}$$

allows Eq. (27) to be further reduced:

$$D = \|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right) t_p}{c\, t_p} + \varepsilon_s \tag{30}$$

$$= \|\mathbf{P}_r^{E_s}(t_s) - \mathbf{P}_s^{E_s}(t_s)\| + \frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_s}(t_s)\right)}{c} + \varepsilon_s \tag{31}$$

$$D = D_s + \varepsilon_s. \tag{32}$$

This shows that the geometric distance D is approximately equal to the quantity D_s defined in Eq. (7) with approximation error ε_s. In GNSS applications, the Sagnac correction is utilized in the form of D_s as shown in Eq. (7), where the error term ε_s is dropped because it is assumed to be negligible relative to the other measurement errors.

To quantify a bound on the error ε_s, we consider each term separately. Using the numeric values stated in Footnote 2, $\|\mathbf{Y}\| \leq 43.24$ m. The first two terms in Eq. (28) are analyzed as follows:

$$\left|\frac{\mathbf{X} \cdot \boldsymbol{\varepsilon}_1}{\|\mathbf{X}\|}\right| = \left|\frac{\mathbf{X}}{\|\mathbf{X}\|} \cdot \boldsymbol{\varepsilon}_1\right| \leq |\boldsymbol{\varepsilon}_1| \leq 1.46 \times 10^{-4} \text{ m;} \tag{33}$$

$$\varepsilon_2 = \frac{1}{2}\frac{\|\mathbf{Y}\|^2}{\|\mathbf{X}\|} \leq \frac{1}{2}\frac{43.24^2}{\|\mathbf{X}\|_{\min}} = 4.54 \times 10^{-5} \text{ m.} \tag{34}$$

At any elevation angle,

$$\left\|\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right\| \leq \left\|\mathbf{P}_s^I(t_s)\right\| \left\|\mathbf{P}_r^I(t_s)\right\| \leq 1.728 \times 10^{14} \text{ m.}$$

The factor $\left|c\, t_p - \|\mathbf{X}\|\right|$ represents the difference between the geometric distance $D = c\, t_p$ for a signal traveling in a vacuum[4] and $\|\mathbf{X}\| = \left\|\mathbf{P}_r^I(t_s) - \mathbf{P}_s^I(t_s)\right\|$. Therefore, $\left|c\, t_p - \|\mathbf{X}\|\right| = |D - \|\mathbf{X}\||$ and Eq. (26) yields:

$$\begin{aligned}\left|c\, t_p - \|\mathbf{X}\|\right| &= \frac{\mathbf{X} \cdot \mathbf{Y}}{\|\mathbf{X}\|} + \frac{1}{2}\frac{\|\mathbf{Y}\|^2}{\|\mathbf{X}\|} \\ &\leq \|\mathbf{Y}\| + \frac{1}{2}\frac{\|\mathbf{Y}\|^2}{\|\mathbf{X}\|_{\min}} \leq 43.24 \text{ m} + 4.54 \times 10^{-5} \text{ m.}\end{aligned}$$

This difference is largest when the receiver and satellite are on the equatorial plane with the satellite at zero degree elevation angle. Combining the above values, the bound on ε_3 is

$$\begin{aligned}|\varepsilon_3| &= \left|\frac{\boldsymbol{\omega}_e \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_r^I(t_s)\right)}{c \|\mathbf{X}\|} \left(c\, t_p - \|\mathbf{X}\|\right)\right| \\ &\leq 8.83 \times 10^{-5} \text{ m.}\end{aligned} \tag{35}$$

Bringing together the three bounds from Eqs. (33)–(35) yields

$$|\varepsilon_s| \leq 3.25 \times 10^{-4} \text{ m.} \tag{36}$$

5.2 Derivation of D_r

This section derives Eq. (8) from Eq. (1). This derivation applies regardless of whether the receiver is stationary or not with respect to the Earth.

Using Eq. (18) to eliminate $\mathbf{P}_r^I(t_r)$ from Eq. (1) yields,

$$D = \|\mathbf{R}_z(-\theta)\, \mathbf{P}_r^I(t_s) + \Delta\mathbf{P}_{r,r_0}^I - \mathbf{P}_s^I(t_s)\|. \tag{37}$$

[3] The upper bound $\|\mathbf{X}\| \leq \|\mathbf{P}_r^I(t_s)\| + \|\mathbf{P}_s^I(t_s)\|$ is valid, but not sufficiently tight. In satellite applications, the elevation must be greater than zero for the satellite signal to reach the receiver. Therefore, a tighter upper bound is $\|\mathbf{X}\| \leq \sqrt{\|\mathbf{P}_s^I(t_s)\|^2 + \|\mathbf{P}_r^I(t_s)\|^2}$.

[4] The Sagnac formulae for the geometric range in Eqs. (7)–(8) are only concerned with signal travel in a vacuum. The delays due to travel through the atmosphere are accounted for separately. Even if these delays were accounted for here, their magnitude in meters is small (i.e., less than 10^2 m.) compared to $c\|\mathbf{X}\|$; therefore, they would not significantly affect the error bound in Eq. (36).

Equation (37) can be written as

$$D = \|\mathbf{A}_1 + \mathbf{B}_1\| = \sqrt{\|\mathbf{A}_1\|^2 + 2\,\mathbf{A}_1 \cdot \mathbf{B}_1 + \|\mathbf{B}_1\|^2} \tag{38}$$

where $\mathbf{A}_1 = \mathbf{P}_r^I(t_s) + \boldsymbol{\Omega}\,\mathbf{P}_r^I(t_s)\,t_p + \Delta\mathbf{P}_{r,r_0}^I - \mathbf{P}_s^I(t_s)$ and $\mathbf{B}_1 = \frac{1}{2}\boldsymbol{\Omega}^2\,\mathbf{P}_r^I(t_s)\,t_p^2$. Applying Taylor's theorem, as expressed in Appendix C, to Eq. (38) with $u = 2\,\mathbf{A}_1 \cdot \mathbf{B}_1 + \|\mathbf{B}_1\|^2$, the first-order approximation yields

$$D = \|\mathbf{A}_1\| + \epsilon_1 + H.O.T \tag{39}$$

where

$$\epsilon_1 = \frac{1}{2\|\mathbf{A}_1\|}(2\,\mathbf{A}_1 \cdot \mathbf{B}_1 + \|\mathbf{B}_1\|^2). \tag{40}$$

Based on the parameter values stated in Footnote 2 and Appendix D, the terms D and $\mathbf{A}_1$ have orders of magnitude of 10^7 meters. The $H.O.Ts$ are on the order of 10^{-15} meters; therefore, the $H.O.Ts$ are negligible are dropped (here and subsequently) from this analysis. The error term ϵ_1 is evaluated in Eq. (D31) of Appendix D.

Using Eq. (B19) from Appendix B, the term $\mathbf{A}_1$ can be rewritten as

$$\begin{aligned}
\mathbf{A}_1 &= \mathbf{P}_r^I(t_s) + \left(\boldsymbol{\omega}_e \times \mathbf{P}_r^I(t_s)\right) t_p + \Delta\mathbf{P}_{r,r_0}^I - \mathbf{P}_s^I(t_s) && (41)\\
&= \left(\mathbf{P}_r^{E_r}(t_r) - \mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) \\
&\quad + \boldsymbol{\omega}_e \times \left(\mathbf{P}_r^{E_r}(t_r) - \mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p + \Delta\mathbf{P}_{r,r_0}^I - \mathbf{P}_s^I(t_s) && (42)\\
&= \mathbf{P}_r^{E_r}(t_r) + \left(\boldsymbol{\omega}_e \times \mathbf{P}_r^{E_r}(t_r)\right) t_p - \mathbf{P}_s^I(t_s) \\
&\quad - \mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I - \boldsymbol{\omega}_e \times \left(\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p + \Delta\mathbf{P}_{r,r_0}^I && (43)\\
&= \mathbf{P}_1 + (\boldsymbol{\omega}_e \times \mathbf{P}_1)\, t_p - \mathbf{P}_s^I(t_s) + \boldsymbol{\Lambda} && (44)
\end{aligned}$$

where $\mathbf{P}_1 = \mathbf{P}_r^{E_r}(t_r)$ and

$$\boldsymbol{\Lambda} = -\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I - \boldsymbol{\omega}_e \times \left(\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p + \Delta\mathbf{P}_{r,r_0}^I. \tag{45}$$

The norm of $\mathbf{P}_1$ has an order of magnitude of 10^6 meters, as stated in Footnote 2. The norm of the perturbation, $\boldsymbol{\Lambda}$, is less than 4.29×10^{-4} meters, as shown in Eq. (D32) of Appendix D.

Let $\mathbf{A}_2 = \mathbf{P}_1 + (\boldsymbol{\omega}_e \times \mathbf{P}_1)\, t_p - \mathbf{P}_s^I(t_s)$, the geometric range D can be written as

$$\begin{aligned}
D &= \|\mathbf{A}_1\| + \epsilon_1 \\
&= \|\mathbf{A}_2 + \boldsymbol{\Lambda}\| + \epsilon_1 \\
&= \sqrt{\|\mathbf{A}_2\|^2 + 2\,\mathbf{A}_2 \cdot \boldsymbol{\Lambda} + \|\boldsymbol{\Lambda}\|^2} + \epsilon_1 && (46)
\end{aligned}$$

where $\|\mathbf{A}_2\|$ has an order of magnitude of 10^7 meters.

Applying Taylor's theorem, as discussed in Appendix C, to Eq. (46) with $u = 2\,\mathbf{A}_2 \cdot \boldsymbol{\Lambda} + \|\boldsymbol{\Lambda}\|^2$, the first-order approximation yields

$$D = \|\mathbf{A}_2\| + \epsilon_1 + \epsilon_2 + H.O.T \tag{47}$$

where $\epsilon_2 = \frac{1}{2\|\mathbf{A}_2\|}\left(2\,\mathbf{A}_2 \cdot \boldsymbol{\Lambda} + \|\boldsymbol{\Lambda}\|^2\right)$. The error term ϵ_2 is evaluated in Eq. (D34) of Appendix D. The $H.O.Ts$ are on the order of 10^{-15} meters; therefore, the $H.O.Ts$ are negligible are dropped from this analysis.

Let $\mathbf{A}_3 = \mathbf{P}_1 - \mathbf{P}_s^I(t_s)$ and $\mathbf{B}_2 = (\boldsymbol{\omega}_e \times \mathbf{P}_1)\, t_p$ yields

$$\begin{aligned}
D &= \|\mathbf{A}_2\| + \epsilon_2 + \epsilon_1 \\
&= \|\mathbf{A}_3 + \mathbf{B}_2\| + \epsilon_2 + \epsilon_1 && (48)\\
&= \sqrt{\|\mathbf{A}_3\|^2 + 2\,\mathbf{A}_3 \cdot \mathbf{B}_2 + \|\mathbf{B}_2\|^2} + \epsilon_2 + \epsilon_1. && (49)
\end{aligned}$$

Applying Taylor's theorem to Eq. (49) as $u = 2\,\mathbf{A}_3 \cdot \mathbf{B}_2 + \|\mathbf{B}_2\|^2$ (See Appendix C), the first-order approximation yields

$$\begin{aligned}
D &= \|\mathbf{A}_3\| + \frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{\|\mathbf{A}_3\|} + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|} + \epsilon_2 + \epsilon_1 + \epsilon_4 && (50)\\
&= \|\mathbf{A}_3\| + \frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{D} + \epsilon_3 + \epsilon_2 + \epsilon_1 + \epsilon_4 && (51)
\end{aligned}$$

where

$$\epsilon_3 = \left(\frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{\|\mathbf{A}_3\|} - \frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{D}\right) + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|}. \tag{52}$$

Based on the analysis in Appendix D, the term $\mathbf{A}_3$ has an order of magnitude of 10^7 meters, while the term $\mathbf{B}_2$ is less than 42.05 meters. Consequently, the error term ϵ_4 represents the $H.O.Ts$ which have an upper bound of 4.75×10^{-5} meters. The error term ϵ_3 is evaluated in Eq. (D36) of Appendix D.

Equation (51) can be further expressed as

$$\begin{aligned}
D &= \|\mathbf{A}_3\| + \frac{t_p}{D}\,(\mathbf{A}_3 \cdot (\boldsymbol{\omega}_e \times \mathbf{P}_1)) + \epsilon_4 + \epsilon_3 + \epsilon_2 + \epsilon_1 \\
&= \|\mathbf{A}_3\| + \frac{t_p}{D}\,(\boldsymbol{\omega}_e \cdot (\mathbf{P}_1 \times \mathbf{A}_3)) + \epsilon_r, && (53)
\end{aligned}$$

where $\epsilon_r = \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4$. For signals traveling in a vacuum so that $D = c\,t_p$ (see Footnote 4):

$$\begin{aligned}
D &= \|\mathbf{A}_3\| + \frac{\boldsymbol{\omega}_e}{c} \cdot (\mathbf{P}_1 \times \mathbf{A}_3) + \epsilon_r \\
&= \|\mathbf{P}_1 - \mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e}{c} \cdot (\mathbf{P}_1 \times \left(\mathbf{P}_1 - \mathbf{P}_s^I(t_s)\right)) + \epsilon_r \\
&= \|\mathbf{P}_1 - \mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^I(t_s) \times \mathbf{P}_1\right) + \epsilon_r
\end{aligned}$$

$$= \|\mathbf{P}_1 - \mathbf{I}\,\mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{I}\,\mathbf{P}_s^I(t_s) \times \mathbf{P}_1\right) + \epsilon_r.$$
$$= \|\mathbf{P}_1 - \mathbf{R}_I^{E_s}\,\mathbf{P}_s^I(t_s)\| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{R}_I^{E_s}\,\mathbf{P}_s^I(t_s) \times \mathbf{P}_1\right) + \epsilon_r.$$
$$= \|\mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_s^{E_s}(t_s)\| + \frac{\boldsymbol{\omega}_e}{c} \cdot \left(\mathbf{P}_s^{E_s}(t_s) \times \mathbf{P}_r^{E_r}(t_r)\right) + \epsilon_r$$
$$D = D_r + \epsilon_r. \tag{54}$$

This shows that the geometric distance D is approximately equal to the quantity D_r defined in Eq. (8) with approximation error ϵ. Based on the analysis in Appendix D, the error is bounded by

$$|\epsilon_r| < 4.54 \times 10^{-5} + 1.36 \times 10^{-4} + 4.29 \times 10^{-4} + 1.46 \times 10^{-4} = 7.57 \times 10^{-4}\ \text{m}. \tag{55}$$

In GNSS applications, the Sagnac correction is utilized in the form of D_r as shown in Eq. (8), where the error term ϵ_r is dropped because it is assumed to be negligible relative to the other measurement errors.

5.3 Addtional notes for D_s and D_r

Although both D_s and D_r do not depend on the satellite or receiver velocity, the error bound for D_r does depend on receiver velocity. As the speed of the receiver increases, the error bound $|\epsilon_2|$ also increases, due to the increased magnitude contribution from the relative movement (see Eqs. (D24) and (D32)). Because D_s is only applicable for stationary receivers, its bound is independent of receiver velocity and will always be smaller that the error bound for D_r. Throughout this article, the upper bound of the Sagnac approximation error refers to the upper bound of D_r.

6 Error bound dependence on orbital radius

The paper has presented a method to bound the approximation error inherent in the Sagnac corrections of Eqs. (7–8). The examples up to this point in the paper have consistently used the parameters stated in Footnote 2 that correspond to one instance of medium earth orbit. Within the wider scope of GNSS, other orbits are also relevant, including low earth orbit, geosynchronous orbit, and many other instances of medium earth orbit. The error bound for any of these orbits can be computed using the same method as that presented in Sec. 5.

Figure 3 displays the Sagnac approximation error bound (for D_r) as a function of the orbital semi-major axis, for values ranging from 6560 km to 45,000 km. The error bound achieves its minimum when the orbit radius is 10,760 km. The region indicated in red represents low earth orbit (radius from 6560 km to 8400 km). The region indicated in yellow represents medium earth orbit (radius from 8400 km to 25,000 km). The cyan region represents geosynchronous orbit (radius at 42,186 km).

For low earth orbits, the approximation error bound increases as the orbit radius decreases, achieving a value of 2.61×10^{-3} m for a radius of 6560 km. For geosynchronous orbits, where the orbit radius is approximately 42,186 km, the approximation error bound is 1.61×10^{-3} m. For most GNSS satellites with medium earth orbits ranging from 25,000 km to 27,000 km, the error bound ranges from 6.71×10^{-4} m to 7.57×10^{-4} m.

7 Conclusion

The Sagnac corrections of Eqs. (7–8) are commonly used in GNSS applications. These corrections are approximations to Eq. (6). This article has, for the first time, analyzed and presented upper bounds on the approximation error. The error bound as a function of orbital radius is displayed in Fig. 3. The bounds derived in this article demonstrate that the upper bound of the approximation error for the geometric range using the Sagnac correction is up to 7.57×10^{-4} m for medium earth orbit GNSS satellites. The maximum error bound increases to 2.6×10^{-3} m for the lowest orbital radius applicable to low earth orbits. For satellites in geosynchronous orbits, the approximation error is smaller than 1.61×10^{-3} m.

Appendix A Notation and reference frames

This appendix discusses notation and useful facts that will be referenced throughout the article.

Useful Fact A.1 *Within this article, without loss of generality, the ECI frame will be defined such that its origin, the origin velocity vector, and the z-axis coincide with those of the ECEF frame at a certain time t_1.*

Let I represents the ECI frame. Let E represents the ECEF frame and $E(t)$ represents the ECEF frame at time t. All frames are right-handed and the sign convention for angles is defined by the right-hand rule.

A.1 Points and frame notation and facts

The symbol $\mathbf{P}_a^{F(t_2)}(t_1)$ represents the coordinate vector of a point $\mathbf{P}_a$ at time t_1 along the axes of frame F at time t_2. When a frame is not rotating, such as the ECI frame, then the '(t_2)' time specification can be dropped. For example, the ECI receiver position coordinate vector at time t can be represented as $\mathbf{P}_a^I(t)$. When a frame, such as the ECEF frame

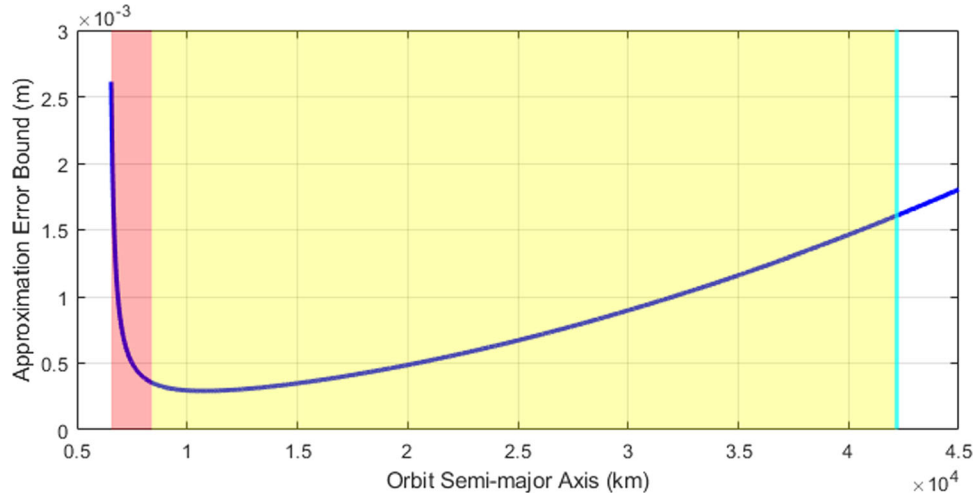

Fig. 3 Sagnac approximation error bounds as a function of orbit semi-major axis length. The red regions represents low earth orbits; the yellow region represents medium earth orbits, and the cyan region represents geosynchronous orbits

is rotating, the '(t_2)' time specification in the frame is critical, because for example a fixed point in the ECI frame will have distinct *coordinates* in the $E(t)$-frames at each t.

Useful Fact A.2 *The position coordinates of a receiver at time t_1 in the ECEF frame at time t_2 are denoted as $\mathbf{P}_r^{E(t_2)}(t_1)$. For convenience in longer derivations, this notation may be simplified as follows:*

$$\mathbf{P}_r^{E_2}(t_1) = \mathbf{P}_r^{E(t_2)}(t_1). \tag{A1}$$

The only situation when the time of the E-frame representation will not be clearly notated is when $t_1 = t_2$:

$$\mathbf{P}_r^{E}(t_1) = \mathbf{P}_r^{E_1}(t_1) = \mathbf{P}_r^{E(t_1)}(t_1) \tag{A2}$$

where $P_r^E(t_1)$ denotes the coordinates of the receiver at time t_1 in the E-frame at t_1 (i.e., the ECEF coordinates of the receiver at t_1).

Useful Fact A.3 *If a receiver is moving relative to the Earth from t_1 to t_2 with Earth-relative velocity $\mathbf{v}^E(t)$, then*

$$\mathbf{P}_r^E(t_2) = \mathbf{P}_r^E(t_1) + \int_{t_1}^{t_2} \mathbf{v}^E(\tau)d\tau. \tag{A3}$$

The relative change of the receiver position is

$$\begin{aligned}\Delta\mathbf{P}^E(t_2, t_1) &= \mathbf{P}_r^{E(t_2)}(t_2) - \mathbf{P}_r^{E(t_1)}(t_1) \\ &= \mathbf{P}_r^E(t_2) - \mathbf{P}_r^E(t_1) = \int_{t_1}^{t_2} \mathbf{v}^E(\tau)d\tau.\end{aligned} \tag{A4}$$

A.2 Rotation notation and facts

The angle from the I-frame to the $E(t)$-frame is

$$\varphi(\delta t) = \alpha + \omega_{ie}\,\delta t. \tag{A5}$$

where α denotes the rotation from the I-frame to the $E(t_1$-frame), $\delta t = t - t_1$, and the Earth rotational rate vector is

$$\boldsymbol{\omega}_e = [0, 0, \omega_{ie}]^\top. \tag{A6}$$

When the E and I frames are aligned at time t_1, as stated in Useful Fact A.1, then $\alpha = 0$. Because the rotation is always around the z-axis, the plane rotation matrix around the z-axis will be used throughout the article. It is defined as (see Eq. (2.26) in (Farrell 2008))

$$\mathbf{R}_z(\varphi) = \begin{bmatrix} \cos(\varphi) & \sin(\varphi) & 0 \\ -\sin(\varphi) & \cos(\varphi) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{A7}$$

Useful Fact A.4 *Consider two Cartesian coordinate frames F_1 and F_2 that share the same origin.*

1. *The coordinates of a point $\mathbf{p}$ in these two frames are related by*

$$\mathbf{P}^{F_2} = \mathbf{R}_{F_1}^{F_2}\,\mathbf{P}^{F_1} \tag{A8}$$

where $\mathbf{R}_{F_1}^{F_2}$ is the rotation matrix from F_1 to F_2. All rotation matrices are orthonormal:

$$\left(\mathbf{R}_{F_1}^{F_2}\right)^{-1} = \left(\mathbf{R}_{F_1}^{F_2}\right)^\top = \mathbf{R}_{F_2}^{F_1}. \tag{A9}$$

2. *In the special case where F_1 rotates about its z-axis with rotation rate vector $\boldsymbol{\omega}_e$ over the time interval from t_1 to t_2 to align its axes with those of F_2, the rotation angle is $\beta = \omega_{ie}(t_2 - t_1)$ and the rotation matrix is*

$$\mathbf{R}_{F_1}^{F_2} = \mathbf{R}_z(\beta). \tag{A10}$$

Appendix A.1 Additional useful facts

Useful Fact A.4 *If a receiver is* Earth-fixed *(i.e., stationary in E-frame), then its Earth-relative velocity is zero. In this special case, the following hold.*

1. *Its ECEF coordinates remain constant: For any t_1 and t_2:*

$$\mathbf{P}_r^E(t_1) = \mathbf{P}_r^{E(t_1)}(t_1) = \mathbf{P}_r^{E(t_2)}(t_2) = \mathbf{P}_r^E(t_2). \tag{A11}$$

2. *Due to the Earth rotation relative to the I-frame, the ECI coordinates are time-varying:*

$$\mathbf{P}_r^I(t_1) = \mathbf{R}_{E(t_1)}^I\,\mathbf{P}_r^E(t_1) \text{ and } \mathbf{P}_r^I(t_2) = \mathbf{R}_{E(t_2)}^I\,\mathbf{P}_r^E(t_2) \tag{A12}$$

3. *While Earth-fixed receiver's ECEF coordinates are constant when the location at a given time is represented in the E-frame at that same time (see Eq. (A11)), the ECEF coordinates will change with time then the ECEF frame time is distinct from the time of the point's location:*

$$\mathbf{P}_r^{E(t_2)}(t_1) = \mathbf{R}_I^{E(t_2)}\,\mathbf{R}_{E(t_1)}^I\,\mathbf{P}_r^{E(t_1)}(t_1).$$

4. *When, for convenience, the I-frame is selected to align with the E-frame at time t_1 (see Useful fact A.1), then:*
   - $\mathbf{R}_{E(t_1)}^I = \mathbf{I}$ *and* $\mathbf{P}_r^I(t_1) = \mathbf{P}_r^E(t_1)$;
   - *Using Eq. (A10) with $F_1 = I$ and $F_2 = E(t_2)$*

$$\begin{aligned}
&\mathbf{R}_I^{E(t_2)} = \mathbf{R}_z(\beta) \text{ and } \mathbf{R}_{E(t_2)}^I = \mathbf{R}_z(-\beta)\\
&\mathbf{P}_r^I(t_2) = \mathbf{R}_{E(t_2)}^I\,\mathbf{P}_r^{E(t_2)}(t_2)\\
&\qquad = \mathbf{R}_z(-\beta)\,\mathbf{P}_r^{E(t_1)}(t_1) \qquad \text{(by eqn. (A11))}\\
&\mathbf{P}_r^I(t_2) = \mathbf{R}_z(-\beta)\,\mathbf{P}_r^I(t_1).
\end{aligned} \tag{A13}$$

Note that Eq. (A13) can be interpreted as rotation of the vector $\mathbf{P}_r^I(t_1)$ through the angle β within the I-frame.

Applying Useful Fact A.4 to the E and I frames with the assumption stated in Useful Fact A.1 for $t_1 = t_s$ (i.e., $\alpha = 0$ in Eq. (A5)), we have the following.

Useful Fact A.5 *A satellite that transmits a signal at time t_s from a fixed position $\mathbf{P}_s(t_s)$ in the I-frame.*

1. *Because the point $\mathbf{P}_s(t_s)$ is fixed in the non-rotating I-frame, the coordinates of this point in the I-frame are fixed: $\mathbf{P}_s^I(t_s)$.*
2. *Due to the Earth rotation relative to the I-frame, the coordinates of the point $\mathbf{P}_s(t_s)$ that is fixed in the I-frame vary with time in the E-frame:*

$$\mathbf{P}_s^{E(t)}(t_s) = \mathbf{R}_I^{E(t)}\mathbf{P}_s^I(t_s),$$

where $\mathbf{R}_I^{E(t)} = \mathbf{R}_z(\omega_{ie}(t - t_s))$.

For example, the position coordinate vector for point $\mathbf{p}_s(t_s)$ in the E-frame at t_r is

$$\mathbf{P}_s^{E(t_r)}(t_s) = \mathbf{R}_I^{E(t_r)}\mathbf{P}_s^I(t_s),$$

which is distinct from its coordinate vector in the E-frame at t_s

$$\mathbf{P}_s^{E(t_s)}(t_s) = \mathbf{R}_I^{E(t_s)}\mathbf{P}_s^I(t_s).$$

Given the Earth rotation vector in Eq. (A6) and that the time-of-signal propagation is

$$t_p = t_r - t_s,$$

the ECEF frame rotates around its z-axis by

$$\theta = \varphi(t_p) = \omega_{ie}\,t_p. \tag{A14}$$

The coordinate vector of a GNSS satellite at t_s in the $E(t_s)$-frame can be computed using the ephemeris (GPS 2013; Glonass 2008; Galileo 2008) and is denoted as $\mathbf{P}_s^{E(t_s)}(t_s)$. Therefore, using Useful Fact A.4, the satellite position in the $E(t_r)$-frame and the $E(t_s)$-frame are related by

$$\mathbf{P}_s^{E(t_r)}(t_s) = \mathbf{R}_{E(t_s)}^{E(t_r)}\,\mathbf{P}_s^{E(t_s)}(t_s). \tag{A15}$$

where, by use of Eqs. (A9–A10),

$$\mathbf{R}_{E(t_s)}^{E(t_r)} = \mathbf{R}_z(\theta) \tag{A16}$$

$$\mathbf{R}_{E(t_r)}^{E(t_s)} = \left(\mathbf{R}_{E(t_s)}^{E(t_r)}\right)^\top = \mathbf{R}_z(\theta)^\top = \mathbf{R}_z(-\theta). \tag{A17}$$

Figure 1 depicts the ECEF frame x and y axes at times $t_r > t_s$, illustrating that the coordinates of points $\mathbf{P}_r(t_r)$ and $\mathbf{P}_s(t_s)$ that are fixed in I-frame will have distinct coordinates in the E-frames defined at different times.

Appendix B Relationships between $\mathbf{P}_r^{E_r}(t_r)$ and $\mathbf{P}_r^I(t_s)$

The purpose of this appendix is to establish the relationship between the symbols $\mathbf{P}_r^{E_r}(t_r)$ and $\mathbf{P}_r^I(t_s)$. This relationship is used in the derivation of Sagnac in Sect. 5.2.

The relationship between $\mathbf{P}_r^{E_r}(t_r)$ and $\mathbf{P}_r^I(t_s)$ is established as follows:

$$\begin{aligned}
\mathbf{P}_r^{E_r}(t_r) &= \mathbf{P}_r^{E_s}(t_s) + \mathbf{P}_r^{E_r}(t_r) - \mathbf{P}_r^{E_s}(t_s)\\
&= \mathbf{P}_r^{E_s}(t_s) + \Delta\mathbf{P}^E\\
&= \mathbf{R}_I^{E_s}\,\mathbf{P}_r^I(t_s) + \Delta\mathbf{P}^E.
\end{aligned} \tag{B18}$$

Substituting Eq. (20) into Eq. (B18) yields

$$\mathbf{P}_r^{E_r}(t_r) = \mathbf{R}_I^{E_s}\,\mathbf{P}_r^I(t_s) + \mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I.$$

Using the fact that yields $\mathbf{R}_I^{E_s} = \mathbf{I}$ yields

$$\mathbf{P}_r^{E_r}(t_r) = \mathbf{P}_r^I(t_s) + \mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I \tag{B19}$$

Appendix C Taylor series approximations

The following Taylor expansions are used in the paper.

C.1 Expansion of $\mathbf{R_z}(\theta)$

Sections 5.1 and 5.2 use the Taylor's series expansion of $\mathbf{R}_z(\theta)$ at $\theta = 0$. The second-order approximation is

$$\mathbf{R}_z(\theta) \approx \mathbf{I} - \mathbf{\Omega}\,t_p + \frac{1}{2}\mathbf{\Omega}^2 t_p^2 = \begin{bmatrix} 1-\frac{\theta^2}{2} & \theta & 0 \\ -\theta & 1-\frac{\theta^2}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ and} \tag{C20}$$

$$\mathbf{R}_z(-\theta) \approx \mathbf{I} + \mathbf{\Omega}\,t_p + \frac{1}{2}\mathbf{\Omega}^2 t_p^2. \tag{C21}$$

where $\mathbf{I}$ is a 3×3 identity matrix and $\mathbf{\Omega} = [\boldsymbol{\omega}_e\times]$ is the cross-product matrix. The rotation angle $\theta = \omega_{ie}\,t_p$, which is defined in Eq. (A14), has a magnitude of about 10^{-5} radians ($\omega_{ie} \approx 7.3\times 10^{-5}\,rad/s$ and $t_p \in (0.06, 0.09)$ seconds). A second-order approximation is used because the matrix is multiplied by $\mathbf{P}_r^I(t_s)$, which has magnitude on the orders of 10^6 meters. The error in dropping terms higher than second-order is less than 10^{-16} meters and is ignored in subsequent analysis.

C.2 Expansion to $\|\mathbf{X}+\mathbf{Y}\|$

Section 5.1 and the derivation from Eq. (38)–Eq. (39) employs a series expansion of $\|\mathbf{X}+\mathbf{Y}\|$. This appendix derives this expansion.

Given two vectors of compatible dimensions $\mathbf{X}$ and $\mathbf{Y}$, the norm $\|\mathbf{X}+\mathbf{Y}\|$ can be manipulated as

$$\|\mathbf{X}+\mathbf{Y}\| = \sqrt{\mathbf{X}\cdot\mathbf{X} + 2\,\mathbf{X}\cdot\mathbf{Y} + \mathbf{Y}\cdot\mathbf{Y}}.$$

Let $u = 2\,\mathbf{X}\cdot\mathbf{Y} + \mathbf{Y}\cdot\mathbf{Y}$ and $f_{\mathbf{X}}(u) = \sqrt{\mathbf{X}\cdot\mathbf{X} + u}$. The Taylor series expansion of $f_{\mathbf{X}}(u)$ around $u = 0$ is

$$f_{\mathbf{X}}(u) = \|\mathbf{X}\| + \frac{u}{2\,\|\mathbf{X}\|} - \frac{u^2}{8\,\|\mathbf{X}\|^3} + \ldots. \tag{C22}$$

Therefore,

$$\|\mathbf{X}+\mathbf{Y}\| = \|\mathbf{X}\| + \frac{u}{2\,\|\mathbf{X}\|} - \frac{u^2}{8\,\|\mathbf{X}\|^3} + \ldots. \tag{C23}$$

Appendix D Bound calculation

This appendix calculates the error term bounds defined in Sect. 5.2.

For subsonic moving receivers, such as automobiles and airplanes, the magnitude of the relative position change $\Delta\mathbf{P}_{r,r_0}^I$ satisfies the following bound

$$\|\Delta\mathbf{P}_{r,r_0}^I\| \le (343\text{ m/s})\,(\bar{t}_p\text{ s}) = 31.75\text{ m}, \tag{D24}$$

because their speed is less than the speed of sound (i.e., 343 m/s) and the maximum value of t_p is $\bar{t}_p = \frac{\|\mathbf{X}\|_{max}}{c}$, as stated in Footnote 2.

The bounds for $\mathbf{A}_1$, $\mathbf{B}_1$ and $\mathbf{B}_2$ defined in Sect. 5.2 are calculated as follows:

$$\begin{aligned}\|\mathbf{A}_1\| &= \|\mathbf{P}_r^I(t_s) + \mathbf{\Omega}\,\mathbf{P}_r^I(t_s)\,t_p + \Delta\mathbf{P}_{r,r_0}^I - \mathbf{P}_s^I(t_s)\| \\ &\le \sqrt{\left(\|\mathbf{P}_r^I(t_s)\| + |\omega_{ie}|\,\|\mathbf{P}_r^I(t_s)\|\,t_p + \|\Delta\mathbf{P}_{r,r_0}^I\|\right)^2 + \|\mathbf{P}_s^I(t_s)\|^2} \\ &\le \sqrt{\left(6.4\times10^6 + 43.42 + 31.75\right)^2 + \left(27\times10^6\right)^2} \\ &= 27.75\times10^6\text{ m},\end{aligned} \tag{D25}$$

$$\begin{aligned}\|\mathbf{A}_1\| &\ge \|\mathbf{P}_s^I(t_s)\| - (\|\mathbf{P}_r^I(t_s)\| + |\omega_{ie}|\,\|\mathbf{P}_r^I(t_s)\|\,t_p + \|\Delta\mathbf{P}_{r,r_0}^I\|) \\ &\ge 27\times10^6 - (6.4\times10^6 + 43.42 + 31.75) \\ &= 20.60\times10^6\text{ m},\end{aligned} \tag{D26}$$

$$\|\mathbf{B}_1\| = \left\|\frac{1}{2}\mathbf{\Omega}^2\,\mathbf{P}_r^I(t_s)\,t_p^2\right\| \le \frac{1}{2}\omega_{ie}^2|\mathbf{P}_r^I(t_s)|\,t_p^2 \le 1.46\times10^{-4}\text{ m} \tag{D27}$$

$$\begin{aligned}\|\mathbf{B}_2\| &= \|\boldsymbol{\omega}_e\times\mathbf{P}_1\,t_p\| \le |\omega_{ie}|\,\|\mathbf{P}_1\|\,|t_p| \\ &= |\omega_{ie}|\,\|\mathbf{P}_r^{E_r}(t_r)\|\,|t_p| = 43.24\text{ m}.\end{aligned} \tag{D28}$$

The calculations for the bounds of $\mathbf{A}_2$ and $\mathbf{A}_3$ follow a similar approach to those of $\mathbf{A}_1$ in Eq. (D25) and Eq. (D26), as described above. Therefore, the bounds of $\mathbf{A}_2$ and $\mathbf{A}_3$ are

$$\|\mathbf{A}_2\| \le \sqrt{\left(\|\mathbf{P}_r^{E_r}(t_r)\| + |\omega_{ie}|\,\|\mathbf{P}_r^{E_r}(t_r)\|\,t_p\right)^2 + \|\mathbf{P}_s^I(t_s)\|^2} \tag{D29}$$

$$\|\mathbf{A}_2\| \ge \|\mathbf{P}_s^I(t_s)\| - \left(\|\mathbf{P}_r^{E_r}(t_r)\| + |\omega_{ie}|\,\|\mathbf{P}_r^{E_r}(t_r)\|\,t_p\right) \tag{D30}$$

Using the numeric values from Footnote 2 yields:

$$20.60\times10^6\text{ m} \le \|\mathbf{A}_2\| \le 27.75\times10^6\text{ m}.$$

Similarly,

$$\|\mathbf{P}_s^I(t_s)\| - \|\mathbf{P}_r^{E_r}(t_r)\| \le \|\mathbf{A}_3\| \le \sqrt{\|\mathbf{P}_r^{E_r}(t_r)\|^2 + \|\mathbf{P}_s^I(t_s)\|^2}$$
$$\Rightarrow 20.60 \times 10^6 \text{ m} \le \|\mathbf{A}_3\| \le 27.75 \times 10^6 \text{ m}$$

Therefore, the bound of the error term ϵ_1 is

$$\begin{aligned} |\epsilon_1| &\le \frac{1}{2\|\mathbf{A}_1\|}(\|\mathbf{B}_1\|^2 + 2\,\|\mathbf{A}_1\|\,\|\mathbf{B}_1\|) \\ &= \frac{\|\mathbf{B}_1\|^2}{2\|\mathbf{A}_1\|} + \|\mathbf{B}_1\| \\ &\le 1.46 \times 10^{-4} \text{ m}. \end{aligned} \tag{D31}$$

The bound of $\mathbf{\Lambda}$ is computed by

$$\begin{aligned} \mathbf{\Lambda} &= -\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I - \boldsymbol{\omega}_e \times \left(\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p + \Delta\mathbf{P}_{r,r_0}^I \\ &= (\mathbf{I} - \mathbf{R}_z(\theta))\,\Delta\mathbf{P}_{r,r_0}^I - \boldsymbol{\omega}_e \times \left(\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p \\ &\approx (\mathbf{\Omega}\, t_p - \frac{1}{2}\mathbf{\Omega}^2 t_p^2)\,\Delta\mathbf{P}_{r,r_0}^I - \boldsymbol{\omega}_e \times \left(\mathbf{R}_z(\theta)\Delta\mathbf{P}_{r,r_0}^I\right) t_p \\ \|\mathbf{\Lambda}\| &\le (\omega_{ie}\, t_p + \frac{1}{2}\omega_{ie}^2 t_p^2)\,\|\Delta\mathbf{P}_{r,r_0}^I\| + \omega_{ie}\,\|\Delta\mathbf{P}_{r,r_0}^I\|\, t_p \\ &\le 4.29 \times 10^{-4} \text{ m}. \end{aligned} \tag{D32}$$

Hence, the bound of the error term ϵ_2 is

$$|\epsilon_2| \le \frac{1}{2\,\|\mathbf{A}_2\|}\left(2\,\|\mathbf{A}_2\|\,\|\mathbf{\Lambda}\| + \|\mathbf{\Lambda}\|^2\right) \tag{D33}$$

$$\begin{aligned} &= \frac{\|\mathbf{\Lambda}\|^2}{2\|\mathbf{A}_2\|} + \|\mathbf{\Lambda}\| \\ &\le 4.29 \times 10^{-4} \text{ m}. \end{aligned} \tag{D34}$$

Using Eqn. (48), the bound of the difference between D and $\|\mathbf{A}_3\|$ is

$$\begin{aligned} |D - \|\mathbf{A}_3\|| &= |\|\mathbf{A}_3 + \mathbf{B}_2\| + \epsilon_2 + \epsilon_1 - \|\mathbf{A}_3\|| \\ &\le |\|\mathbf{B}_2\| + \epsilon_2 + \epsilon_1| \\ &\le 43.24 \text{ m}. \end{aligned} \tag{D35}$$

Therefore, the bound of the error term ϵ_3 is computed by

$$\begin{aligned} \epsilon_3 &= \left(\frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{\|\mathbf{A}_3\|} - \frac{\mathbf{A}_3 \cdot \mathbf{B}_2}{D}\right) + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|} \\ &= \frac{\left|D - \|\mathbf{A}_3\|\right|}{\|\mathbf{A}_3\|D}\,(\mathbf{A}_3 \cdot \mathbf{B}_2) + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|} \\ &\le \frac{\left|D - \|\mathbf{A}_3\|\right|\|\mathbf{A}_3\|\|\mathbf{B}_2\|}{\|\mathbf{A}_3\|D} + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|} \\ &= \frac{\left|D - \|\mathbf{A}_3\|\right|\|\mathbf{B}_2\|}{D} + \frac{\|\mathbf{B}_2\|^2}{2\,\|\mathbf{A}_3\|} \\ &\le 9.08 \times 10^{-5} + 4.54 \times 10^{-5} \\ &= 1.36 \times 10^{-4} \text{ m}. \end{aligned} \tag{D36}$$

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References

Ashby N (1994) Relativity in the future of engineering. IEEE Trans Instrum Measure 43(4):505–514

Ashby N (1995) Relativistic effects in the global positioning system. J Syst Eng Electron 6(4):199–237

Ashby N (1999) Global positioning system receivers and relativity. Technical Note 1385, National Institute of Standards and Technology

Ashby N (2004) The Sagnac effect in the global positioning system. Springer, Dordrecht

BeiDou (2018) BeiDou navigation satellite system open service performance standard (Version 2.0). China Satellite Navigation Office

Bidikar B, Rao GS, Ganesh L (2016) Sagnac effect and set error based pseudorange modeling for GPS applications. Procedia Comp Sci 87:172–177

Farrell J (2008) Aided navigation: GPS with high rate sensors. McGraw-Hill, Inc

Galileo (2008) Galileo signal in space interface control document (OS SIS ICD). Galileo Open Service

Galileo (2019) European GNSS (Galileo) open service service definition document. European GNSS Service Centre

Gift SJ (2014) Time transfer and the Sagnac correction in the GPS. Appl Phys Res 6(6):1

Glonass IC (2008) Glonass interface control document. Russian Institute of Space Device Engineering, Moscow, Russia

GPS (2013) Global positioning systems directorate system engineering & integration interface specification IS-GPS-200H. Navstar GPS Space Segment/Navigation User Interfaces

GPS Product Team (2020) Global positioning system standard positioning service performance analysis report. GPS Product Team, Washington, DC, USA

Groves PD (2013) Principles of GNSS, inertial, and multisensor integrated navigation systems. Artech House

Hu W, Farrell JA (2019) Derivation of earth-rotation correction (sagnac) and analysis of the effect of receiver clock bias. Bourns College of Engineering, UC Riverside

Hu W, Neupane A, Farrell JA (2022) Using PPP information to implement a global real-time virtual network DGNSS approach. IEEE Trans Veh Technol 71(10):10337–10349

Leick A, Rapoport L, Tatarnikov D (2015) GPS satellite surveying. John Wiley & Sons

Misra P, Enge P (2011) Global positioning system: signals, measurements, and performance, 2nd edn. Ganga-Jamuna Press, Lincoln, MA

Post EJ (1967) Sagnac effect. Rev Modern Phys 39(2):475

Su CC (2001) Reinterpretation of the Michelson–Morley experiment based on the GPS Sagnac correction. In: APMC 2001. 2001 Asia-Pacific microwave conference (Cat. No.01TH8577), vol 2, pp. 468–471

Takasu T (2013) RTKLIB ver. 2.4. 2 Manual. RTKLIB: An open source program package for GNSS Positioning

Teunissen PJ, Montenbruck O (2017) Springer Handbook of Global Navigation Satellite Systems, vol 10. Springer